

Functional & Performance Testing Template

Model Performance Test

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	

Functional Testing

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	User Input Validation	Enter valid and invalid Life Expectancy, Education, and Income values	Valid inputs accepted and invalid inputs rejected with validation messages	Validation worked correctly	Pass
FT-02	Numeric Input Validation	Enter alphabets, special characters, empty values, and negative values	System displays appropriate validation errors	Validation displayed successfully	Pass
FT-03	HDI Prediction	Enter all valid development indicators and click Predict	Machine Learning model predicts HDI score successfully	HDI predicted correctly	Pass
FT-04	Model Loading	Start Flask application and load the saved HDI.pkl model	Model loads successfully without errors	Model loaded successfully	Pass
FT-05	Data Preprocessing Validation	Submit records containing missing values or incomplete fields	Missing values handled or appropriate error messages displayed	Preprocessing completed successfully	Pass

FT-06	HDI Category Classification	Predict HDI for different input combinations	System correctly classifies Low, Medium, High, and Very High HDI	Classification accurate	Pass
FT-07	Result Display Verification	Submit valid input and view prediction page	HDI score, category, and indices displayed correctly	Results displayed correctly	Pass
FT-08	Database Storage Verification	Submit prediction and verify SQLite database	Prediction details stored successfully	Data stored successfully	Pass
FT-09	Prediction History	Make multiple predictions and open History page	Previous predictions displayed correctly	History loaded successfully	Pass
FT-10	Report Generation	Generate prediction report after successful prediction	SVG report generated and saved successfully	Report generated successfully	Pass

Performance Testing

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
PT-01	Prediction Response Time	Submit prediction request and measure response time	Response generated within 2 seconds	Average response time: 1.3 seconds	Pass
PT-02	Concurrent Prediction Requests	Submit multiple prediction requests simultaneously	System handles concurrent requests successfully	Requests processed correctly	Pass
PT-03	Dataset Processing	Train ML model using complete HDI dataset	Training completes successfully	Training completed successfully	Pass
PT-04	Model Prediction Speed	Predict HDI for 100 different records	Predictions generated without delay	Average prediction time: 0.02	Pass

				seconds per record	
PT-05	Database Performance	Store and retrieve prediction history repeatedly	Database operations complete without delay	Database responded normally	Pass
PT-06	Application Load Test	Access application from multiple browsers	Application remains stable and responsive	Stable performance observed	Pass
PT-07	Memory Usage Test	Run application continuously for several hours	No abnormal memory increase or crash	Memory usage remained stable	Pass
PT-08	Model Loading Performance	Restart application multiple times	Model loads within acceptable time	Model loaded in approximately 0.8 seconds	Pass